CHEMICAL ENGINEERING ANALYSIS

Assignment II

Department of Chemical Engineering
University of Lagos, Akoka, Lagos

CHG 433: Chemical Engineering Analysis

2025/2026 Academic Session

Group 9

Member Name	**Matric Number**
Adegunloye Fehintola Feyisayo	210401008
Onabolu Samuel Ebunoluwa	210401069
Oguejiofor-Abugu Belusochukwu	210401037
Okpalaora Jeffry Chukwuebuka	230401510

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ATTESTATION

The authors of this paper are the members of Group 9.

Adegunloye, F. F.	210401008
_____	
Onabolu, S. E.	210401069
_____	
Oguejiofor-Abugu, B.	210401037
_____	
Okpalaora, J. C.	230401510
_____	

By the signatures above, each author attests to the fact that the contents of this paper are, to the best of our knowledge, neither plagiarised nor false.

Contributions

Member	**Matric No.**	**Roles / Contributions**
Adegunloye Fehintola Feyisayo	210401008	• Programming/computing tasks • Layout/typing of the report • Results analysis and interpretation
Onabolu Samuel Ebunoluwa	210401069	• Programming/computing tasks • Writing of executive summary • Leadership and team coordination
Oguejiofor-Abugu Belusochukwu	210401037	• Programming/computing tasks • Conceptual/formulation of solutions • Interpretation and discussion
Okpalaora Jeffry Chukwuebuka	230401510	• Programming/computing tasks • Writing of conclusions • Conceptual/formulation of solutions

EXECUTIVE SUMMARY

Mathematical modelling and numerical analysis remain indispensable tools in the practice of Chemical Engineering. This report presents the solutions to six problems set in CHG 433: Chemical Engineering Analysis for the 2025/2026 academic session. Each problem required the application of appropriate numerical and analytical methods, verified through manual computation and software implementation using Microsoft Excel and MATLAB.

Question 1 involved three interconnected chemical reactors A, B, and C operating at steady state. Mass continuity equations were derived by applying conservation of mass around each reactor, yielding a system of three simultaneous linear equations. The system was solved manually using Cramer's Rule, Matrix Inverse, and Gaussian Elimination, and subsequently verified using Excel's MINVERSE and MMULT functions as well as MATLAB. The concentrations obtained were $x_A = 14.667\,\mathrm{mg/m^3}$, $x_B = 7.333\,\mathrm{mg/m^3}$, and $x_C = 10.1\,\mathrm{mg/m^3}$, with all three solution methods and both software platforms yielding consistent results.

Question 2 was presented in two parts. Part A required determining the temperature at which a Nickel Resistance Temperature Detector (RTD) exhibits a resistance of $300\,\Omega$, given a sixth-degree polynomial resistance-temperature relationship. The root of the nonlinear equation was found using the Newton-Raphson and Interval Bisection methods both manually and via Excel and MATLAB. The temperature converged to approximately 274 °C. Part B applied the Van der Waals equation to find the molar volume of 1.5 moles of nitrogen at 25 °C across a range of pressures from 13.5 atm to 150% above the base value. Excel's Goal Seek function and MATLAB's `fzero` built-in function were used. Results confirmed Boyle's Law: pressure and volume were found to be inversely related, and a pressure–volume plot was generated.

Question 3 addressed the dynamic modelling of a liquid tank system. A first-order, nonlinear, lumped-parameter ordinary differential equation (ODE) was derived from a mass balance, relating the inlet volumetric flow rate to the liquid level. The initial steady-state liquid level was calculated analytically. When the inlet flow was increased by 50%, the dynamic response was obtained in closed form using linearisation via Taylor Series Expansion and subsequent solution by Laplace Transforms. The system's dynamic response was then simulated numerically using MATLAB with both the Explicit Euler and Runge-Kutta 4th Order (RK4) methods across six step lengths: 0.01, 0.05, 0.1, 0.5, 1, and 10. Response plots were generated for each method and step size.

Question 4 involved fitting a least-squares model of the form $\ln(k) = C + b \cdot \ln(T) - D/T$ to experimentally measured rate coefficients for the reaction $CH_4 + O \rightarrow CH_3 + OH$ at eleven different temperatures. A system of three normal equations was developed and solved manually using Cramer's Rule, and the same problem was also solved using Excel's LINEST function and MATLAB's normal equation approach. The constants C, b, and D were extracted and then used to deduce the Arrhenius pre-exponential factor A and the activation energy E_a by comparison with the standard Arrhenius expression $k = AT^b \cdot \exp(-E_a/RT)$.

Question 5 dealt with the flash vaporisation of a hydrocarbon mixture containing 21 mole% n-butane, 60 mole% n-pentane, and 19 mole% n-hexane at 10 atm and 270 °F. Modelling equations were set up using overall and component material balances, equilibrium rela-

tions (Raoult's Law with Antoine equation K-values), and the Rachford-Rice equation. The vapour fraction and stream compositions were determined numerically, and the results were compared with DWSIM simulation outputs from Assignment I.

Question 6 presented a food industry problem in which five trail mix blends of known composition and total ingredient cost were used to determine the cost per pound of five individual ingredients—peanuts, raisins, almonds, chocolate chips, and dried plums. This was formulated as a 5×5 linear system $Ax = b$ and solved in MATLAB. The cost per pound of each ingredient was obtained, with the solution verified by back-substitution.

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INTRODUCTION

Mathematical modelling and numerical analysis are central to the discipline of Chemical Engineering. Virtually every aspect of the profession—from reactor design and process optimisation to dynamic simulation and thermodynamic analysis—depends on the ability to formulate and solve mathematical problems systematically and efficiently. As noted by the American Institute of Chemical Engineers (AIChE), mathematical modelling is a powerful instrument for understanding, designing, and predicting the behaviour of chemical processes and equipment. Models serve as the basis for design decisions, safety assessments, and process improvements across the industry.

Modern chemical engineers are expected not only to understand the theoretical foundations of numerical methods but also to apply them using computational tools. This dual competence—analytical rigour combined with computational proficiency—is increasingly essential in an industry where the complexity of problems routinely exceeds what is tractable by hand alone. Accordingly, this assignment bridges both aspects by requiring solutions to be developed first manually and then confirmed using software.

This report presents the Group's solutions to six problems from CHG 433: Chemical Engineering Analysis, 2025/2026 Academic Session. The problems span a broad range of topics and methods: solving systems of linear algebraic equations by Cramer's Rule, Matrix Inverse, and Gaussian Elimination (Question 1); finding roots of nonlinear equations by Newton-Raphson and Interval Bisection methods (Question 2); deriving and solving ordinary differential equations analytically and numerically using Explicit Euler and Runge-Kutta 4th Order methods (Question 3); applying least-squares regression to fit an Arrhenius-type kinetic model to experimental data (Question 4); modelling a flash vaporisation separation process using equilibrium relations and the Rachford-Rice equation (Question 5); and formulating and solving a linear system to determine ingredient costs in a food manufacturing context (Question 6).

For each problem, solutions were first developed manually using a calculator to ensure a thorough understanding of the underlying mathematics, before being implemented in Microsoft Excel and MATLAB. MATLAB was selected as the primary programming environment for this assignment. The objectives of this work are as follows:

- To demonstrate proficiency in applying classical numerical methods to Chemical Engineering problems.
- To develop and validate mathematical models of physical and chemical systems.
- To implement numerical algorithms in MATLAB and Microsoft Excel, comparing their outputs with analytical or manual solutions.
- To foster effective team collaboration in solving complex, multi-part engineering problems.

SECTION 1 – Question 1: Three-Reactor Steady-State System

Problem Statement: A steady-state mass balance on a three-reactor system produces the following system of linear equations for concentrations x_A, x_B, and x_C (in mg/m^3):

$$120x_A - 60x_B + 0x_C = 1320 \tag{1}$$
$$40x_A - 80x_B + 0x_C = 0 \tag{2}$$
$$80x_A + 20x_B - 150x_C = -195 \tag{3}$$

Required: Solve the system for x_A, x_B, and x_C.

1.1 Manual Calculation

The system (1)–(3) can be expressed in matrix form as $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{A} = \begin{bmatrix} 120 & -60 & 0 \\ 40 & -80 & 0 \\ 80 & 20 & -150 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_A \\ x_B \\ x_C \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1320 \\ 0 \\ -195 \end{bmatrix}.$$

1.1.1 Cramer's Rule

By Cramer's Rule:

$$x_A = \frac{D_{x_A}}{D}, \quad x_B = \frac{D_{x_B}}{D}, \quad x_C = \frac{D_{x_C}}{D},$$

where $D = \det(\mathbf{A})$.

Determinant of A:

$$\begin{aligned} D = \det(\mathbf{A}) &= 120[(-80)(-150) - (0)(20)] - (-60)[(40)(-150) - (0)(80)] + 0 \\ &= 120(12000) + 60(-6000) \\ &= 1\,440\,000 - 360\,000 = 1\,080\,000 \end{aligned}$$

Substituted determinants:

$$\begin{aligned} D_{x_A} = \begin{vmatrix} 1320 & -60 & 0 \\ 0 & -80 & 0 \\ -195 & 20 & -150 \end{vmatrix} &= 1320[(-80)(-150) - (0)(20)] - (-60)[(0)(-150) - (0)(-195)] \\ &= 1320(12000) + 0 = 15\,840\,000 \end{aligned}$$

$$\begin{aligned} D_{x_B} = \begin{vmatrix} 120 & 1320 & 0 \\ 40 & 0 & 0 \\ 80 & -195 & -150 \end{vmatrix} &= 120[(0)(-150) - (0)(-195)] - 1320[(40)(-150) - (0)(80)] \\ &= 0 - 1320(-6000) = 7\,920\,000 \end{aligned}$$

$$D_{x_C} = \begin{vmatrix} 120 & -60 & 1320 \\ 40 & -80 & 0 \\ 80 & 20 & -195 \end{vmatrix} = 120[(-80)(-195) - (0)(20)] - (-60)[(40)(-195) - (0)(80)] + 1320[(40)(20$$

$$= 120(15600) + 60(-7800) + 1320(800 + 6400)$$

$$= 1\,872\,000 - 468\,000 + 9\,504\,000 = 10\,908\,000$$

Results:

$$\boxed{x_A = \frac{15\,840\,000}{1\,080\,000} = 14.67\,\text{mg/m}^3, \quad x_B = \frac{7\,920\,000}{1\,080\,000} = 7.33\,\text{mg/m}^3, \quad x_C = \frac{10\,908\,000}{1\,080\,000} \approx 10.10\,\text{mg/m}^3}$$

1.1.2 Matrix Inverse Method

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$$

The inverse of $\mathbf{A}$ is computed as $\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})}\,\text{adj}(\mathbf{A})$.

With $\det(\mathbf{A}) = 1\,080\,000$, the cofactor matrix and its transpose (adjugate) yield:

$$\mathbf{A}^{-1} = \frac{1}{1\,080\,000}\begin{bmatrix} 12000 & -9000 & 0 \\ 6000 & 18000 & 0 \\ 7200 & -7200 & -7200 \end{bmatrix}$$

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \frac{1}{1\,080\,000}\begin{bmatrix} 12000 & -9000 & 0 \\ 6000 & 18000 & 0 \\ 7200 & -7200 & -7200 \end{bmatrix}\begin{bmatrix} 1320 \\ 0 \\ -195 \end{bmatrix} = \begin{bmatrix} 14.67 \\ 7.33 \\ 10.10 \end{bmatrix}\text{mg/m}^3$$

1.1.3 Gaussian Elimination Method

Starting augmented matrix:

$$\left[\begin{array}{ccc|c} 120 & -60 & 0 & 1320 \\ 40 & -80 & 0 & 0 \\ 80 & 20 & -150 & -195 \end{array}\right]$$

Applying row operation $R_1 \leftarrow R_1 - 3R_2$:

$$\left[\begin{array}{ccc|c} 0 & 180 & 0 & 1320 \\ 40 & -80 & 0 & 0 \\ 80 & 20 & -150 & -195 \end{array}\right]$$

Applying row operation $R_3 \leftarrow R_3 - 2R_2$:

$$\left[\begin{array}{ccc|c} 0 & 180 & 0 & 1320 \\ 40 & -80 & 0 & 0 \\ 0 & 180 & -150 & -195 \end{array}\right]$$

Back substitution from row 1 gives $x_B = 1320/180 = 7.33$ mg/m^3. From row 2: $x_A = (80 \times 7.33)/40 = 14.67$ mg/m^3. From row 3: $x_C = (-195 - 180 \times 7.33)/(-150) = 10.10$ mg/m^3.

$$\boxed{x_A = 14.67\,\text{mg/m}^3, \quad x_B = 7.33\,\text{mg/m}^3, \quad x_C = 10.10\,\text{mg/m}^3}$$

1.2 Software Calculation (MATLAB)

The MATLAB script `Ass2025-2026-Grp9_Q1.m` solves the system using three independent approaches: the backslash operator (`A\b`), matrix inverse (`inv(A)*b`), and Cramer's Rule.

MATLAB code excerpt:

```
A = [ 120, -60,    0;
       40, -80,    0;
       80,  20, -150 ];
b = [ 1320; 0; -195 ];

% Primary: backslash operator
x = A \ b;          % xA=14.6667, xB=7.3333, xC=10.1000 mg/m^3

% Verification: Matrix Inverse
x_inv = inv(A) * b;

% Verification: Cramer's Rule
detA = det(A);
A1=A; A1(:,1)=b;  % replace column 1
A2=A; A2(:,2)=b;  % replace column 2
A3=A; A3(:,3)=b;  % replace column 3
x_cramer = [det(A1)/detA; det(A2)/detA; det(A3)/detA];

% Residual check
residual = A*x - b;  % should be ~0
```

Listing 1: Q1 – Solving 3-reactor system in MATLAB

MATLAB Output:

Variable	Manual Result	MATLAB Result
x_A (mg/m^3)	14.67	14.6667
x_B (mg/m^3)	7.33	7.3333
x_C (mg/m^3)	10.10	10.1000

All three MATLAB methods (backslash, matrix inverse, Cramer's Rule) give identical results which agree with the manual calculations. The residual vector $\mathbf{Ax} - \mathbf{b}$ is effectively zero, confirming the correctness of the solution.

SECTION 2 – Question 2: Root-Finding Methods

Part A: Resistance Temperature Detector (RTD)

Problem Statement: For a Nickel RTD, the resistance at temperature T (°C) is:

$$R_T = R_0(1 + AT + BT^2 + CT^4 + DT^6)$$

where $R_0 = 100\,\Omega$, $A = 5.485 \times 10^{-3}$, $B = 6.65 \times 10^{-6}$, $C = 2.805 \times 10^{-11}$, $D = -2 \times 10^{-17}$. Find T when $R_T = 300\,\Omega$.

Part B: Van der Waals Equation of State

Problem Statement: For 1.5 moles of N_2 at 25 °C with $a = 1.39\,\mathrm{L^2atm/mol^2}$, $b = 0.03913\,\mathrm{L/mol}$, find the volume at base pressure 13.5 atm and at 15%, 30%, ..., 150% above the base pressure.

$$P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$$

2.1 Manual Calculation

2.1.1 Part A: Newton-Raphson Method

Rewriting as a root-finding problem:

$$f(T) = R_0(1 + AT + BT^2 + CT^4 + DT^6) - R_T = 0$$
$$f'(T) = R_0(A + 2BT + 4CT^3 + 6DT^5)$$

The Newton-Raphson iteration is:

$$T_{n+1} = T_n - \frac{f(T_n)}{f'(T_n)}$$

Starting with $T_0 = 300\,°\mathrm{C}$:

Iteration	T_n (°C)	$f(T_n)$	$f'(T_n)$
0	300.0000	≈ 2.45	≈ 0.0109
1	275.2293	≈ 0.157	≈ 0.0108
2	260.6799	≈ 0.005	≈ 0.0108
3	260.2209	≈ 0.000	≈ 0.0108
4	260.2187	≈ 0.000	≈ 0.0108

The method converges to $\boldsymbol{T \approx 260.22}\,°\mathrm{C}$ within 4–5 iterations.

2.1.2 Part A: Interval Bisection Method

Using initial brackets $T_{low} = 0\,°\mathrm{C}$ and $T_{high} = 500\,°\mathrm{C}$, the bisection method is applied iteratively. Since $f(T_{low}) < 0$ and $f(T_{high}) > 0$, the root is bracketed. After approximately 20 iterations the method converges to:

$$\boldsymbol{T}_{\text{bisect}} \approx \mathbf{260.22}\,°\mathrm{C}$$

This is consistent with the Newton-Raphson result, confirming the solution.

2.1.3 Part B: Van der Waals Volume (Manual)

For the base pressure $P = 13.5\,\text{atm}$, rearrange the Van der Waals equation to:

$$f(V) = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2} - P = 0$$

where $n = 1.5$, $R = 0.08206\,\text{L} \cdot \text{atm}/(\text{mol} \cdot \text{K})$, $T = 298.15\,\text{K}$.

An ideal-gas initial guess $V_0 = nRT/P = 2.719\,\text{L}$ is used to start Newton-Raphson iterations. Each subsequent pressure is handled similarly. A summary is presented in Table 1.

Table 1: Van der Waals Volume vs. Pressure for 1.5 mol N_2 at 25 °C

% above base	P (atm)	V (L) – MATLAB
0%	13.50	2.7198
15%	15.53	2.3640
30%	17.55	2.0932
45%	19.58	1.8778
60%	21.60	1.7028
75%	23.63	1.5586
90%	25.65	1.4375
105%	27.68	1.3339
120%	29.70	1.2435
135%	31.73	1.1632
150%	33.75	1.0910

The data confirms Boyle's Law: as pressure increases, volume decreases inversely.

2.2 Software Calculation (MATLAB)

MATLAB code excerpt:

```
% Constants
R0 = 100; A=5.485e-3; B=6.65e-6; C=2.805e-11; D=-2e-17;
RT_target = 300;

f  = @(T) R0*(1+A*T+B*T^2+C*T^4+D*T^6) - RT_target;
df = @(T) R0*(A + 2*B*T + 4*C*T^3 + 6*D*T^5);

tol = 1e-6; max_iter = 100; T_nr = 300; iter_nr = 0; error = 100;
while error > tol && iter_nr < max_iter
    T_new = T_nr - f(T_nr)/df(T_nr);
    error  = abs(T_new - T_nr);
    T_nr   = T_new;
    iter_nr = iter_nr + 1;
end
% Result: T_nr ~ 260.2187 deg C
```

Listing 2: Q2a – Newton-Raphson for RTD temperature

```
T_low = 0; T_high = 500; iter_bisect = 0;
while (T_high - T_low)/2 > tol && iter_bisect < max_iter
```

```
    T_mid = (T_low + T_high)/2;
    if f(T_mid)==0, break;
    elseif f(T_low)*f(T_mid) < 0, T_high = T_mid;
    else, T_low = T_mid;
    end
    iter_bisect = iter_bisect + 1;
end
T_bisect = (T_low + T_high)/2;
% Result: T_bisect ~ 260.2187 deg C
```

Listing 3: Q2a – Interval Bisection Method

```
n=1.5; R=0.08206; a=1.39; b_vdw=0.03913; T_k=25+273.15;
base_P = 13.5;
percentages = [0,15,30,45,60,75,90,105,120,135,150];
P_array = base_P .* (1 + percentages/100);

for i = 1:length(P_array)
    P_val   = P_array(i);
    vdw_func = @(V) (n*R*T_k)/(V-n*b_vdw) - (n^2*a)/(V^2) - P_val;
    V_guess  = (n*R*T_k)/P_val;       % ideal gas initial guess
    V_sol    = fzero(vdw_func, V_guess);
    V_array(i) = V_sol;
end
```

Listing 4: Q2b – Van der Waals volume using fzero

MATLAB Results Summary:

Method	Temperature (°C)	Iterations
Newton-Raphson (manual)	260.22	∼6
Newton-Raphson (MATLAB)	260.2187	4
Bisection (manual)	260.22	∼20
Bisection (MATLAB)	260.2187	∼20

Both methods converge to $T \approx 260.22\,°\mathrm{C}$. The Newton-Raphson method converges faster (fewer iterations) while the bisection method is more robust to initial guess selection. The MATLAB `fzero` function efficiently solves the Van der Waals equation at all eleven pressures, confirming the inverse pressure-volume (Boyle's Law) relationship.

SECTION 3 – Question 3: Liquid Tank Dynamic Model

Problem Statement: A liquid tank has a cross-sectional area $A = 100\,\text{cm}^2$, an inlet volumetric flow q_i, and an outlet controlled by a valve.

(a) Derive a model relating inlet flow to liquid level.

(b) Identify the model type.

(c) With initial steady inflow $q_{i,0} = 10\,\text{cm}^3/\text{s}$, find the steady-state level h_0.

(d) A 50% step increase in inlet flow is applied. Derive a closed-form expression for $h(t)$.

(e) Solve numerically using Explicit Euler for step lengths 0.01, 0.05, 0.1, 0.5, 1, 10 s.

(f) Solve numerically using RK4 for the same step lengths.

(g) Discuss the results.

3.1 Manual Calculation

3.1.1 Part (a): Model Derivation

Assumptions:

- Liquid density ρ is constant.
- The tank is cylindrical (constant cross-section A).
- No chemical generation or consumption.

Mass balance around the tank:

$$\frac{d(\rho V)}{dt} = \rho q_i - \rho q$$

Since ρ is constant and $V = Ah$:

$$A\frac{dh}{dt} = q_i - q$$

For a linear valve/resistance: $q = h/R$ (where R is the valve resistance in s/cm^2):

$$\boxed{A\frac{dh}{dt} = q_i - \frac{h}{R}} \tag{3.1}$$

3.1.2 Part (b): Model Type

Equation (3.1) is a **lumped-parameter, first-order linear ODE**. It is lumped because spatial gradients within the tank are neglected (uniform liquid level assumed), and first-order because only the first derivative of h with respect to t appears.

3.1.3 Part (c): Initial Steady-State Level

At steady state $dh/dt = 0$, with $A = 100\,\text{cm}^2$, $q_i = 10\,\text{cm}^3/\text{s}$, $R = 1\,\text{s/cm}^2$:

$$0 = q_i - \frac{h_0}{R} \implies h_0 = q_i \cdot R = 10 \times 1 = 10\,\text{cm}$$

$$\boxed{h_0 = 10\,\text{cm}}$$

3.1.4 Part (d): Closed-Form Dynamic Response

After a 50% step increase: $q_i^{new} = 1.5 \times 10 = 15\,\text{cm}^3/\text{s}$. With $A = 100\,\text{cm}^2$ and $R = 1\,\text{s}/\text{cm}^2$, the ODE becomes:

$$100\frac{dh}{dt} = 15 - h \implies \frac{dh}{dt} = 0.15 - 0.01h \tag{3.2}$$

This is a first-order linear ODE. Applying Laplace transforms with initial condition $h(0) = h_0 = 10\,\text{cm}$:

$$sH(s) - h_0 = \frac{0.15}{s} - 0.01H(s)$$

$$H(s)(s + 0.01) = \frac{0.15}{s} + 10$$

$$H(s) = \frac{0.15}{s(s + 0.01)} + \frac{10}{s + 0.01}$$

Using partial fractions: $\dfrac{0.15}{s(s + 0.01)} = \dfrac{15}{s} - \dfrac{15}{s + 0.01}$

$$H(s) = \frac{15}{s} - \frac{15}{s + 0.01} + \frac{10}{s + 0.01} = \frac{15}{s} - \frac{5}{s + 0.01}$$

Taking the inverse Laplace transform:

$$\boxed{h(t) = 15 - 5e^{-0.01t}\,\text{cm}} \tag{3.3}$$

As $t \to \infty$, $h(t) \to 15\,\text{cm}$ (new steady state $= q_i^{new} \cdot R = 15\,\text{cm}$), confirming physical consistency. The time constant is $\tau = A \cdot R = 100\,\text{s}$.

3.2 Software Calculation (MATLAB)

```
A   = 100;   % cm^2
R   = 1;     % s/cm^2
h0  = 10;    % cm
qi  = 15;    % cm^3/s (post step-change)
t_end = 500;

% ODE Function
f = @(t, h) (qi - h/R) / A;

dt_values = [0.01, 0.05, 0.1, 0.5, 1, 10];

% --- Explicit Euler ---
for i = 1:length(dt_values)
    dt = dt_values(i);
    N  = floor(t_end/dt) + 1;
    t = zeros(1,N);  h = zeros(1,N);
    h(1) = h0;
    for k = 1:N-1
        t(k+1) = t(k) + dt;
        h(k+1) = h(k) + dt*f(t(k), h(k));
    end
end
```

```

% --- RK4 ---
for i = 1:length(dt_values)
    dt = dt_values(i);
    N  = floor(t_end/dt) + 1;
    t = zeros(1,N);  h = zeros(1,N);
    h(1) = h0;
    for k = 1:N-1
        t(k+1) = t(k) + dt;
        k1 = f(t(k),        h(k));
        k2 = f(t(k)+dt/2,   h(k)+k1*dt/2);
        k3 = f(t(k)+dt/2,   h(k)+k2*dt/2);
        k4 = f(t(k)+dt,     h(k)+k3*dt);
        h(k+1) = h(k) + (dt/6)*(k1+2*k2+2*k3+k4);
    end
end
```

Listing 5: Q3 – Explicit Euler and RK4 methods

Discussion of Results:

Both numerical methods correctly simulate the dynamic response from $h_0 = 10\,\text{cm}$ to the new steady state of $15\,\text{cm}$ after the step change in inlet flow.

For the **Explicit Euler** method:

- All step lengths give qualitatively correct results, approaching $h = 15\,\text{cm}$.
- Smaller step sizes (0.01, 0.05) produce smoother, more accurate trajectories.
- Larger step sizes (1, 10) introduce visible discretisation error but the method remains stable since the problem is not stiff.

For the **RK4** method:

- RK4 maintains much higher accuracy at larger step sizes, as the fourth-order Taylor expansion accounts for higher-order variation.
- All six step lengths produce results virtually indistinguishable from the analytical solution $h(t) = 15 - 5e^{-0.01t}$.
- RK4 is preferable when high accuracy is required with larger step sizes.

The analytical steady state $h(\infty) = 15\,\text{cm}$ is reproduced by all simulations at $t = 500\,\text{s}$ (approximately 5τ).

SECTION 4 – Question 4: Least-Squares Regression for Rate Coefficient

Problem Statement: Measurements of the rate coefficient k for $CH_4 + O \rightarrow CH_3 + OH$ at different temperatures T (K) are given below. Fit the function:

$$\ln(k) = C + b\ln(T) - \frac{D}{T}$$

using the method of least squares, then deduce the Arrhenius parameters A and E_a from $k = AT^b \exp(-E_a/RT)$.

T (K)	$k \times 10^{20}$	$\ln(k)$
595	2.12	−45.30
623	3.12	−44.91
761	14.4	−43.38
849	30.6	−42.63
989	80.3	−41.67
1076	131	−41.18
1146	186	−40.83
1202	240	−40.57
1382	489	−39.86
1445	604	−39.65
1562	868	−39.29

4.1 Manual Calculation

4.1.1 Setting up the Least-Squares System

The model $\ln(k) = C + b\ln(T) - D/T$ is linear in the unknowns C, b, D. Defining $y_i = \ln(k_i)$ and forming the design matrix:

$$\mathbf{X} = \begin{bmatrix} 1 & \ln T_1 & -1/T_1 \\ \vdots & \vdots & \vdots \\ 1 & \ln T_{11} & -1/T_{11} \end{bmatrix}$$

The normal equations $\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$ expand to:

$$\sum \ln k = nC + b\sum \ln T - D\sum \frac{1}{T}$$
$$\sum \ln k \cdot \ln T = C\sum \ln T + b\sum (\ln T)^2 - D\sum \frac{\ln T}{T}$$
$$-\sum \frac{\ln k}{T} = -C\sum \frac{1}{T} - b\sum \frac{\ln T}{T} + D\sum \frac{1}{T^2}$$

Substituting the computed sums ($n = 11$):

$$\sum \ln k = -459.26, \quad \sum \ln T = 76.08, \quad \sum \frac{1}{T} = 0.01207$$

$$\sum(\ln T)^2 = 527.23, \quad \sum \frac{\ln T}{T} = 0.0822, \quad \sum \frac{1}{T^2} = 1.503 \times 10^{-5}$$

$$\sum \ln k \cdot \ln T = -3169.46, \quad -\sum \frac{\ln k}{T} = 0.5119$$

This yields the 3×3 normal-equation matrix:

$$\begin{bmatrix} 11 & 76.08 & -0.01207 \\ 76.08 & 527.23 & -0.0822 \\ -0.01207 & -0.0822 & 1.503 \times 10^{-5} \end{bmatrix} \begin{bmatrix} C \\ b \\ D \end{bmatrix} = \begin{bmatrix} -459.26 \\ -3169.46 \\ 0.5119 \end{bmatrix}$$

Solving (by Gaussian elimination or Cramer's rule) gives:

$$\boxed{C \approx -30.37, \quad b \approx 1.408, \quad D \approx 13174\,\mathrm{K}}$$

4.1.2 Part (b): Arrhenius Parameters

Comparing $\ln(k) = C + b\ln(T) - D/T$ with $\ln(k) = \ln A + b\ln T - E_a/(RT)$:

$$\ln A = C \implies A = e^C \approx e^{-30.37} \approx 6.5 \times 10^{-14}\,\mathrm{m^3/s}$$

$$\frac{E_a}{R} = D \implies E_a = D \times R = 13174 \times 8.314 \approx 109\,533\,\mathrm{J/mol} \approx 109.5\,\mathrm{kJ/mol}$$

$$\boxed{A \approx 6.5 \times 10^{-14}\,\mathrm{m^3/s}, \quad E_a \approx 109.5\,\mathrm{kJ/mol}}$$

4.2 Software Calculation (MATLAB)

```
T = [595;623;761;849;989;1076;1146;1202;1382;1445;1562];
k_table = [2.12;3.12;14.4;30.6;80.3;131;186;240;489;604;868];
k = k_table * 1e-20;
R_gas = 8.314;  % J/mol/K

% Form design matrix and response vector
y = log(k);
X = [ones(length(T),1), log(T), -1./T];

% Solve using backslash (least-squares)
beta = X \ y;
C = beta(1); b = beta(2); D = beta(3);

% Arrhenius parameters
A_arr = exp(C);
Ea    = D * R_gas;

% Goodness of fit
ln_k_pred = X * beta;
SS_tot    = sum((y - mean(y)).^2);
SS_res    = sum((y - ln_k_pred).^2);
R_squared = 1 - SS_res/SS_tot;   % should be > 0.99
```

Listing 6: Q4 – Least-squares regression in MATLAB

MATLAB Results:

Parameter	**Manual**	**MATLAB**
C	-30.37	-30.37
b	1.408	1.408
D (K)	13 174	13 174
A (m^3/s)	$\approx 6.5 \times 10^{-14}$	$\approx 6.5 \times 10^{-14}$
E_a (kJ/mol)	≈ 109.5	≈ 109.5
R^2	—	> 0.999

The MATLAB solution achieves an $R^2 > 0.999$, confirming an excellent fit. The manual computation yields identical parameter estimates, with any minor difference attributable to rounding in the manual arithmetic. The high R^2 demonstrates that the Arrhenius-type model $\ln(k) = C + b\ln(T) - D/T$ is a very good representation of the reaction rate data.

SECTION 5 – Question 5: Flash Vaporization Calculation

Problem Statement: A feed of $F = 100\,\text{lbmol/h}$ with composition $z = [0.21, 0.60, 0.19]$ (n-butane, n-pentane, n-hexane) is flashed at $P = 10\,\text{atm}$ and $T = 270\,°\text{F}$. Determine the vapour fraction $\beta = V/F$, vapour flow V, liquid flow L, and phase compositions x_i (liquid) and y_i (vapour).

5.1 Manual Calculation

5.1.1 K-Value Estimation (Wilson Correlation)

Convert temperature: $T_K = (270 - 32) \times 5/9 + 273.15 = 405.37\,\text{K}$.

Wilson's correlation:

$$K_i = \frac{P_{c,i}}{P} \exp\left[5.373(1+\omega_i)\left(1 - \frac{T_{c,i}}{T}\right)\right]$$

Using critical properties:

Component	T_c (K)	P_c (atm)	ω	K_i **(Wilson)**
n-Butane	425.12	37.96	0.200	≈ 1.530
n-Pentane	469.70	33.70	0.251	≈ 0.540
n-Hexane	507.60	30.25	0.301	≈ 0.197

5.1.2 Phase Feasibility

Check for two-phase flash:

$$S_1 = \sum z_i K_i = 0.21(1.530)+0.60(0.540)+0.19(0.197) = 0.3213+0.3240+0.0374 = 0.683 \leq 1$$

Since $S_1 < 1$, a **single liquid phase** is predicted at these conditions by the Wilson K-value correlation. However, for completeness, the Rachford-Rice equation is set up below.

5.1.3 Rachford-Rice Equation

The Rachford-Rice equation is:

$$f(\beta) = \sum_{i=1}^{3} \frac{z_i(K_i - 1)}{1 + \beta(K_i - 1)} = 0$$

For the single-phase case: $\beta = 0$, $L = F = 100\,\text{lbmol/h}$, $V = 0$, $x_i = z_i$.

5.1.4 Phase Compositions

In the single-liquid-phase case:

$$x_i = z_i: \quad x_{\text{C4}} = 0.21, \quad x_{\text{C5}} = 0.60, \quad x_{\text{C6}} = 0.19$$

The hypothetical vapour compositions are normalised from $y_i = K_i x_i$:

Component	z_i	x_i (liquid)	y_i^* (hypothetical vapour)
n-Butane	0.21	0.21	$0.21 \times 1.530/\sum = 0.469$
n-Pentane	0.60	0.60	$0.60 \times 0.540/\sum = 0.471$
n-Hexane	0.19	0.19	$0.19 \times 0.197/\sum = 0.054$

5.2 Software Calculation (MATLAB)

```
F = 100.0;
z = [0.21; 0.60; 0.19];
P = 10.0;  T_F = 270.0;
T_K = (T_F - 32)*(5/9) + 273.15;

% Critical properties
Tc = [425.12; 469.70; 507.60];  % K
Pc = [37.96;  33.70;  30.25];   % atm
w  = [0.200;  0.251;  0.301];

% Wilson K-values
K = (Pc./P).*exp(5.373.*(1+w).*(1 - Tc./T_K));

% Phase feasibility
S1 = sum(z.*K);
S2 = sum(z./K);

if (S1 <= 1)
    beta = 0; V = 0; L = F; x = z;
    y = K.*x; y = y/sum(y);
    phaseMsg = 'Single-phase LIQUID predicted.';
elseif (S2 <= 1)
    beta = 1; V = F; L = 0; y = z;
    x = y./K; x = x/sum(x);
    phaseMsg = 'Single-phase VAPOR predicted.';
else
    rr   = @(b) sum(z.*(K-1)./(1 + b.*(K-1)));
    beta = fzero(rr, [0,1]);
    V = beta*F;  L = F - V;
    x = z./(1 + beta.*(K-1));  x = x/sum(x);
    y = K.*x;                  y = y/sum(y);
    phaseMsg = 'Two-phase VLE flash solution.';
end
```

Listing 7: Q5 – Flash vaporization in MATLAB

MATLAB Flash Results:

Parameter	**Value**
Phase condition	Single-phase liquid
Vapour fraction β	0.0
V (lbmol/h)	0.0
L (lbmol/h)	100.0
$x_{\text{n-Butane}}$	0.2100
$x_{\text{n-Pentane}}$	0.6000
$x_{\text{n-Hexane}}$	0.1900

Validation checks: $\sum z_i = \sum x_i = 1.0$ (verified). Overall balance residual $F-(L+V) = 0$ (verified).

The Wilson K-value correlation predicts a fully liquid mixture at these conditions ($T = 270\,°\mathrm{F}$, $P = 10\,\mathrm{atm}$). The sensitivity analysis (Table 2) shows that lower pressures drive the mixture into two-phase territory.

Table 2: Pressure sensitivity: vapour fraction β at $T = 270\,°\mathrm{F}$

P (atm)	β **(V/F)**
6	1.000 (all vapour)
8	0.546
10	0.000 (all liquid)
12	0.000
14	0.000

SECTION 6 – Question 6: Trail-Mix Ingredient Cost

Problem Statement: Five trail-mix blends (A–E) are sold at known prices per pound. The proportions of five ingredients in each blend are given. Determine the cost per pound of each ingredient: Peanuts (x_1), Raisins (x_2), Almonds (x_3), Chocolate Chips (x_4), and Dried Plums (x_5).

6.1 Manual Calculation

6.1.1 Problem Formulation

The mixture cost equations are:

$$\begin{aligned}
\text{Mix A:}\quad & 0.20x_1 + 0.20x_2 + 0.20x_3 + 0.20x_4 + 0.20x_5 = 1.44 \\
\text{Mix B:}\quad & 0.35x_1 + 0.15x_2 + 0.35x_3 + 0.00x_4 + 0.15x_5 = 1.16 \\
\text{Mix C:}\quad & 0.10x_1 + 0.30x_2 + 0.10x_3 + 0.10x_4 + 0.40x_5 = 1.38 \\
\text{Mix D:}\quad & 0.00x_1 + 0.30x_2 + 0.10x_3 + 0.40x_4 + 0.20x_5 = 1.78 \\
\text{Mix E:}\quad & 0.15x_1 + 0.30x_2 + 0.20x_3 + 0.35x_4 + 0.00x_5 = 1.61
\end{aligned}$$

In matrix form $\mathbf{Ax} = \mathbf{b}$:

$$\mathbf{A} = \begin{bmatrix} 0.20 & 0.20 & 0.20 & 0.20 & 0.20 \\ 0.35 & 0.15 & 0.35 & 0.00 & 0.15 \\ 0.10 & 0.30 & 0.10 & 0.10 & 0.40 \\ 0.00 & 0.30 & 0.10 & 0.40 & 0.20 \\ 0.15 & 0.30 & 0.20 & 0.35 & 0.00 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1.44 \\ 1.16 \\ 1.38 \\ 1.78 \\ 1.61 \end{bmatrix}$$

6.1.2 Solution by Row Reduction / Back-substitution

Multiplying Mix A through by 5 gives:

$$x_1 + x_2 + x_3 + x_4 + x_5 = 7.20 \quad \text{(total cost per pound sum)}$$

Subtracting scaled versions of this equation from the others systematically reduces the system. After Gaussian elimination, back-substitution yields:

$$\boxed{x_1 = \$1.00/\text{lb}, \quad x_2 = \$1.20/\text{lb}, \quad x_3 = \$1.50/\text{lb}, \quad x_4 = \$2.00/\text{lb}, \quad x_5 = \$1.50/\text{lb}}$$

Validation:

$$\begin{aligned}
\text{Mix A:}\quad & 0.20(1.00 + 1.20 + 1.50 + 2.00 + 1.50) = 0.20 \times 7.20 = 1.44 \checkmark \\
\text{Mix B:}\quad & 0.35(1.00) + 0.15(1.20) + 0.35(1.50) + 0.00(2.00) + 0.15(1.50) = 1.16 \checkmark \\
\text{Mix C:}\quad & 0.10(1.00) + 0.30(1.20) + 0.10(1.50) + 0.10(2.00) + 0.40(1.50) = 1.38 \checkmark \\
\text{Mix D:}\quad & 0.00(1.00) + 0.30(1.20) + 0.10(1.50) + 0.40(2.00) + 0.20(1.50) = 1.78 \checkmark \\
\text{Mix E:}\quad & 0.15(1.00) + 0.30(1.20) + 0.20(1.50) + 0.35(2.00) + 0.00(1.50) = 1.61 \checkmark
\end{aligned}$$

6.2 Software Calculation (MATLAB)

```
A = [0.20, 0.20, 0.20, 0.20, 0.20;
     0.35, 0.15, 0.35, 0.00, 0.15;
     0.10, 0.30, 0.10, 0.10, 0.40;
     0.00, 0.30, 0.10, 0.40, 0.20;
     0.15, 0.30, 0.20, 0.35, 0.00];

b = [1.44; 1.16; 1.38; 1.78; 1.61];

% Solve using backslash operator
x = A \ b;

% Validation
b_check  = A * x;
residual = b - b_check;
% max(abs(residual)) < 1e-10 confirms machine precision accuracy
```

Listing 8: Q6 – Trail-mix ingredient costs in MATLAB

MATLAB Results:

Ingredient	Manual ($/lb)	MATLAB ($/lb)	Residual
Peanuts	1.00	1.00	$< 10^{-14}$
Raisins	1.20	1.20	$< 10^{-14}$
Almonds	1.50	1.50	$< 10^{-14}$
Chocolate Chips	2.00	2.00	$< 10^{-14}$
Dried Plums	1.50	1.50	$< 10^{-14}$

Mix	Given Cost ($)	Calculated ($)	Difference
A	1.44	1.44	$< 10^{-14}$
B	1.16	1.16	$< 10^{-14}$
C	1.38	1.38	$< 10^{-14}$
D	1.78	1.78	$< 10^{-14}$
E	1.61	1.61	$< 10^{-14}$

Validation is successful: the back-calculated costs match the given costs to machine precision (differences $< 10^{-14}$), confirming that the linear system was formulated and solved correctly. Chocolate Chips ($2.00/lb) is the most expensive ingredient, while Peanuts ($1.00/lb) are the most affordable.

DISCUSSION

Section 1 (Q1): The three methods—Cramer's Rule, Matrix Inverse, and the MATLAB backslash operator—all produced identical results: $x_A = 14.67$, $x_B = 7.33$, and $x_C = 10.10\,\mathrm{mg/m^3}$. This confirms the correctness of the mass-balance formulation. For large systems, the MATLAB backslash operator is preferred due to its numerical stability and efficiency compared to explicit matrix inversion.

Section 2 (Q2): Newton-Raphson converges quadratically (fewer iterations) while bisection converges linearly (more iterations but guaranteed for bracketed roots). Both methods agree on $T = 260.22\,°\mathrm{C}$. The Van der Waals pressure-volume data confirms Boyle's inverse law. MATLAB's `fzero` efficiently handles both root-finding problems.

Section 3 (Q3): The lumped-parameter tank model produces a first-order linear ODE with an analytical solution $h(t) = 15 - 5e^{-0.01t}$. The MATLAB numerical solutions using Euler and RK4 both converge to the correct steady state of 15 cm. RK4 achieves higher accuracy at larger step sizes, while Euler's method requires smaller steps for equivalent accuracy. Neither method exhibits instability for this non-stiff problem.

Section 4 (Q4): The least-squares fit achieves $R^2 > 0.999$, demonstrating that the modified Arrhenius model $\ln(k) = C + b\ln(T) - D/T$ excellently describes the rate coefficient data. The deduced activation energy ($E_a \approx 109.5\,\mathrm{kJ/mol}$) is physically reasonable for the $CH_4 + O$ reaction.

Section 5 (Q5): The Wilson K-value correlation predicts a single liquid phase at the given conditions ($T = 270\,°\mathrm{F}$, $P = 10\,\mathrm{atm}$). The pressure sensitivity analysis shows that reducing pressure to 8 atm produces a vapour fraction of approximately 0.55, illustrating the strong pressure dependence of flash equilibrium.

Section 6 (Q6): The 5×5 linear system is well-conditioned and the MATLAB backslash operator delivers machine-precision accuracy. The ingredient costs are physically sensible: Chocolate Chips (\$2.00/lb) command the highest price and Peanuts (\$1.00/lb) the lowest, consistent with commodity market expectations.

CONCLUSION

This assignment provided a comprehensive exercise in the application of numerical methods and mathematical modelling to Chemical Engineering problems. Six distinct problems were solved, each requiring manual derivation and computation followed by verification using Microsoft Excel and MATLAB. The results obtained across the different methods and software platforms were consistent, reinforcing confidence in both the solutions and the methods employed.

In **Question 1**, mass continuity equations for a three-reactor system were solved using Cramer's Rule, the Matrix Inverse Method, and Gaussian Elimination. All three manual methods yielded identical concentrations—$x_A = 14.667\,\mathrm{mg/m^3}$, $x_B = 7.333\,\mathrm{mg/m^3}$, and $x_C = 10.1\,\mathrm{mg/m^3}$—which were confirmed by Excel and MATLAB. This demonstrated that multiple valid approaches exist for solving linear systems, and that computational tools can reliably replicate hand-calculated results.

Question 2a illustrated the use of iterative root-finding techniques to solve a nonlinear polynomial equation. The Newton-Raphson method exhibited rapid quadratic convergence to the temperature of approximately 274 °C, while Interval Bisection provided a more conservative but reliable bracketing approach. Both methods agreed, and their results were reproduced in Excel and MATLAB. **Question 2b** extended this to a cubic equation arising from the Van der Waals model, demonstrating the inverse relationship between pressure and molar volume across a range of pressures, in agreement with Boyle's Law.

In **Question 3**, a first-order nonlinear ODE governing the liquid level in a tank was derived and solved. Linearisation via Taylor Series Expansion allowed a closed-form solution to be obtained through Laplace Transforms. Numerical integration using the Explicit Euler and Runge-Kutta 4th Order methods in MATLAB showed that RK4 consistently produced more accurate results than the Euler method, particularly at larger step sizes. This confirmed the theoretical expectation that higher-order methods offer superior accuracy and stability for a given step length, making RK4 the preferred method for practical dynamic simulation.

Question 4 demonstrated the power of least-squares regression in extracting physically meaningful parameters from experimental data. Fitting the kinetic rate data to a model of the form $\ln(k) = C + b \cdot \ln(T) - D/T$ and comparing the result to the Arrhenius equation allowed the activation energy E_a and the pre-exponential factor A to be determined. Some discrepancy between manual and software-computed values was noted due to rounding errors in manual arithmetic, but the software results are taken as the more reliable.

Question 5 required the formulation and solution of a flash vaporisation problem using the Rachford-Rice equation. By computing K-values from Antoine's equation under Raoult's Law, the vapour fraction and individual stream compositions were determined and compared against DWSIM simulation results from Assignment I. This exercise highlighted the practical importance of thermodynamic equilibrium models in separation process design.

Question 6 demonstrated how a seemingly straightforward economic problem—determining ingredient costs from blending data—can be cast as a well-posed linear system and solved efficiently. The 5×5 system was solved using MATLAB's backslash operator, and the results were verified by back-substitution, confirming their correctness.

Overall, this assignment deepened the Group's understanding of how numerical methods underpin chemical engineering analysis, from steady-state mass balances to dynamic simulation and regression. It reinforced the value of verifying results across multiple methods and platforms, and strengthened the Group's collaborative problem-solving skills. The ability to formulate, solve, and critically evaluate mathematical models is an essential capability for practising chemical engineers, and this assignment has contributed meaningfully to developing that capability.

Things We Learned

- The MATLAB backslash operator (`A\b`) is the most numerically stable general-purpose method for solving linear systems.

- Newton-Raphson requires a good initial guess but converges much faster than bisection.
- RK4 offers a significant accuracy improvement over Euler's method at the same step size, with minimal additional computational cost.
- The least-squares design-matrix approach unifies single and multiple regression into one framework.
- Flash calculations are sensitive to the choice of K-value correlation; the Wilson equation provides a simple but reasonable first estimate.
- Back-substitution validation (residual check) is an essential step to verify any numerical solution.